

Scaling Book — Section 1 Exercises

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Q1

Let $X \in \mathbb{R}^{B,D}$, $Y \in \mathbb{R}^{D,F}$, $Z \in \mathbb{R}^{B,F}$, with

$$XY = Z \quad \text{in int8.}$$

(i) Bytes loaded / written

$$\begin{aligned} \text{loaded:} & \quad BD + DF \\ \text{written:} & \quad BF \end{aligned}$$

(ii) FLOPs

There are BF dot products, each of cost $D + D - 1$:

$$\underbrace{BF}_{\text{dot prods}} \cdot \underbrace{(D + D - 1)}_{\text{dot prod cost}} \approx 2BFD.$$

(iii) Arithmetic intensity

$$\text{AI} = \frac{2BFD}{BD + DF + BF}.$$

If $B \ll D, F$:

$$\text{AI} \approx \frac{2BFD}{DF} = 2B.$$

(iv) Lower / upper bound on runtime

$$T_{\text{math}} = \frac{\text{FLOPs}}{\text{FLOPs/s}} = \frac{2BFD}{3.94\text{e}14}, \quad T_{\text{comm}} = \frac{\text{bytes}}{\text{bytes/s}} = \frac{BD + DF + BF}{8.2\text{e}11}.$$

$$\begin{aligned} \text{lower bound:} & \quad \max(T_{\text{math}}, T_{\text{comm}}) \\ \text{upper bound:} & \quad T_{\text{math}} + T_{\text{comm}} \end{aligned}$$

(plug in above)

Q2

$$\text{bf16}[B, D] \cdot \text{int8}[D, F] \longrightarrow \text{bf16}[B, F]$$

$$\text{total FLOPs} = 2BFD$$

$$\text{bytes moved} = 2BD + DF + 2BF$$

$$\text{AI} = \frac{2BFD}{2BD + DF + 2BF} \quad (\text{assume } B \ll F, D) = 2B.$$

At the corner of the roofline model:

$$\text{Intensity}(\text{accel}) = \frac{1.97\text{e14 FLOPs}}{8.2\text{e11 bytes}}.$$

$$2B = \text{Intensity}(\text{accel}) \implies B = \frac{1.97\text{e14}}{2 \cdot 8.2\text{e11}} \approx 120.$$

Q3

Done on computer. Plot peak achievable FLOPs/s as a function of batch size for $F = D = 4096$ and $F = D = 1024$ (TPU v5e: 1.97×10^{14} FLOPs/s, 8.2×10^{11} bytes/s).

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3
4 batches = np.arange(1, 500)
5 BANDWIDTH = 8.2e11
6 FLOPS_S = 1.97e14
7
8
9 def calc_peak_flops(f, d):
10     def solve(b):
11         flops = 2 * b * f * d
12         bytes_moved = (2 * b * d) + (d * f) + (2 * b * f)
13         ai = flops / bytes_moved
14         peak_flops_s = np.minimum(ai * BANDWIDTH, FLOPS_S)
15
16         return peak_flops_s
17
18     return solve
19
20
21 f_d_4096 = calc_peak_flops(4096, 4096)
22 f_d_1024 = calc_peak_flops(1024, 1024)
23
24 plt.plot(batches, f_d_4096(batches))
25 plt.plot(batches, f_d_1024(batches))
26 plt.xlabel("batch size")
27 plt.ylabel("peak flops/s")
28 plt.show()
```

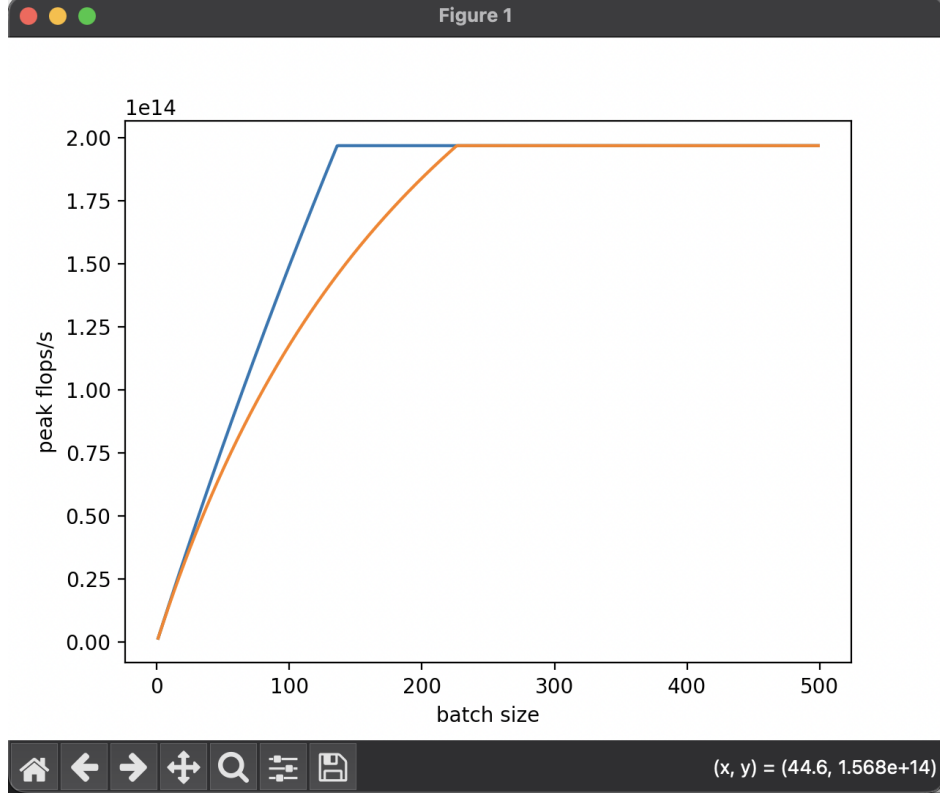


Figure 1: *

Peak FLOPs/s vs. batch size. Blue: $F = D = 4096$. Orange: $F = D = 1024$. Both saturate at the accelerator peak (1.97×10^{14} FLOPs/s); the larger matmul ($D = 4096$) reaches the corner of the roofline at a smaller batch size.

Q4

$$\text{int8}[B, D] \cdot \text{int8}[B, D, F] \longrightarrow \text{int8}[B, F]$$

(Per-batch matmul: for each b , a length- D vector multiplies a $D \times F$ matrix.)

$$\begin{aligned} \text{FLOPs} &= BF(2D) = 2DBF \\ \text{bytes} &= BD + BDF + BF \\ \text{AI} &= \frac{2DBF}{BD + BDF + BF} \end{aligned}$$

Q5

H100 SXM specs:

$$\begin{aligned} \text{bf16: } & \underbrace{1.979 \times 10^{15} \text{ FLOPs/s}}_{\text{w/ sparsity}} \implies \text{true value: } \frac{1.979}{2} \times 10^{15} \text{ FLOPs/s} \\ \text{bw: } & 3.35 \times 10^{12} \text{ bytes/s} \end{aligned}$$

$$\text{AI}(\text{matmul}) = B \quad (\text{assuming small } B).$$

$$B_{\text{compute-bound}} = \left\lfloor \frac{\frac{1.979}{2} \times 10^{15} \text{ FLOPs/s}}{3.35 \times 10^{12} \text{ bytes/s}} \right\rfloor \approx 295 \text{ tokens} \quad (\text{rounding down}).$$